AN2DL - Second Challenge Report
Team Name : Dumpling

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1 Introduction

The objective of this challenge is *multiclass image classification*. Specifically, the task requires **designing a neural network model capable of classifying images into four different molecular subtypes**: Luminal A, Luminal B, HER2(+) and Triple Negative. The provided dataset consists of 691 image/mask pairs for training and 477 pairs for final evaluation, characterized by varying image sizes.

Our team decided to organize the workflow in the following manner:

- First, to **analyze the data** and perform necessary data cleaning;
- **Data Augmentation:** Following the data cleaning phase, we implemented data augmentation techniques. This step was necessary to address the **scarcity of data** and to prevent the model from overfitting on the small training set.
- Then, to develop a **CNN from scratch**, starting from the basics;
- Finally, to proceed towards more advanced models applying **Transfer Learning** and **Fine-Tuning** techniques.

2 Problem Analysis

1. Dataset characteristics:
   Upon a preliminary visual inspection, it became immediately apparent that the dataset contained significant noise and artifacts. Specifically, several images were obstructed by *green blobs*, and in some anomalous cases, the tissue texture was overlaid with images of the character *Shrek*. The training dataset is imbalanced. In fact, the distribution of the molecular subtypes is not uniform, with the Triple-Negative subtype being significantly underrepresented
2. Main challenges:
   - **Data Scarcity (Small Data Problem):** With only 691 training samples we have severe risk of *overfitting*.
   - **Class Imbalance:** The unequal distribution of images across the four molecular subtypes creates a bias.
   - **Noisy Data and Artifacts:** The presence of visual artifacts—such as green blobs and superimposed images (e.g., Shrek)—introduces non-informative noise. These outliers can confuse the network.

- **High Inter-class Similarity:** Images are visually complex. The morphological differences between subtypes (e.g., Luminal A vs. Luminal B) can be subtle and difficult for a model to distinguish.
- Dimensionality : The dataset consists of high-resolution images with variable dimensions.

3. Initial assumptions:
   Although the use of masks was optional, we hypothesized that they play a crucial role. By isolating the discriminative elements, they effectively filter out the background noise, allowing the model to focus only on the relevant tissue.

3 Method

Data processing

- We began by cleaning the images and removing any anomalies (581 samples remaining). This process highlighted a significant class imbalance, particularly regarding the few available Triple Negative examples.
- Instead of resizing the images—which causes a loss of diagnostic detail—we utilized a **patching strategy**. Finally, to guarantee uniform input sizes for the model, we applied zero-padding to the patches where necessary.
- We one hot encoded the labels: Luminal B (0), Luminal A (1), HER2(+) (2), and Triple Negative (3).
- Split the 581 samples into 85% training and 15% validation sets.
- We incorporated the tissue mask as a fourth channel alongside the standard RGB input. This 4-channel configuration guides the model to focus on the tissue regions while preserving all original pixel data.

Model training and optimization

- **Overfitting:** Addressed with Dropout, L2 regularization (λ_2).
- **Class Imbalance:** Managed by balancing the training set, using **loss weighting**, and applying geometric and color augmentation.
- **Performance:** Optimized using early stopping, and label smoothing and MIXUP.
- **Fine Tuning:** We fine-tune a pretrained backbone by freezing early layers and updating only higher-level layers together with the task-specific classification head. This strategy balances generalization and domain adaptation while mitigating overfitting.

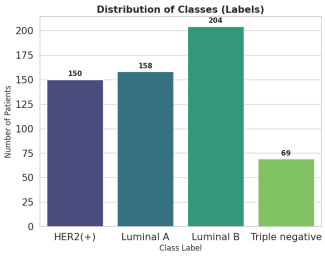

Figure 1: Class distribution after image cleaning

4 Experiments

Our experimentation began with baseline architectures before moving to more complex sequential models. The base models that have been used are as follows:

- **Simple CNN** Firstly, we implemented a lightweight four-layer CNN. We quickly realized that this model was too simple for the task. After applying data augmentation, the model struggled to train and defaulted to guessing the most frequent label (Class 0) for all samples, effectively failing to learn any meaningful patterns. This model achieved a score of 0.28 on the Kaggle test set.
- **VGG16** We then switched to transfer learning with VGG16, which improved our validation accuracy to 44.46% (34% on Kaggle). We observed severe overfitting and very slow training speeds. Consequently, we decided to move to a lighter and more efficient model.
- **RESNET18** Switching to ResNet18 resulted in a promising local validation score of 48.38%. However,no gains on the Kaggle test, where we

Table 1: Best results of each model on the validation set.

Model	Accuracy	Precision	Recall	F1 Score
Simple CNN	0.3750	0.4007	0.3750	0.3824
VGG16	0.4526	0.5054	0.4526	0.4446
ConvNext	0.4530	0.4552	0.4530	0.4460
ResNet18	**0.4872**	**0.4852**	**0.4872**	**0.4838**

achieved 39.10%. Although the reduced complexity helped stabilize training compared to previous ones, we have no better generalization.

- **Another Models** We also evaluated larger models, such as ResNet50, but found them to be unsuitable for our dataset, likely due to limited data availability and increased model complexity leading to stronger overfitting.

5 Results

As shown in Figure 2, the dataset is highly skewed with few Triple Negative samples.

- **Key Achievements:** The **ResNet18 with 4-channel MIL inputs** proved most effective, reaching a validation score of **48.38%** and outperforming the computationally expensive VGG19.
- **Generalization Gap:** Despite this success, the model scored 39.10% on the Kaggle test set, indicating that the test distribution remains challenging despite the focused tissue strategy.

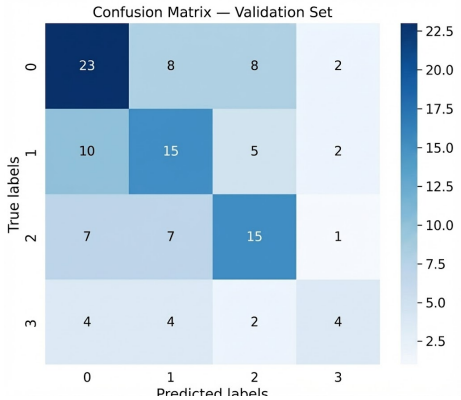

Figure 2: Confusion matrix ResNet18

6 Discussion

- **Class Confusion:** The model frequently confuses biologically similar classes (0 and 1), indicating that the dataset imbalance prevents the learning of subtle distinguishing features.
- **Fragmented Context:** Analyzing isolated patches deprives the model of the global tissue architecture. This lack of a holistic view creates a bottleneck, as the network misses large-scale diagnostic patterns.
- **Data Scarcity:** We failed to effectively resolve the class imbalance, as generating synthetic samples or sourcing compatible external data proved unreliable for this specific domain.

7 Conclusions

- Restate main contributions:
  Every teammate gave their best effort, and everyone worked hard to achieve the best possible results. Each teammate contributed in different ways during the various phases of the project.
- Suggest improvements:
  During our experiments, we observed that heavy models like VGG19 were difficult to train due to hardware limitations. We suggest that future iterations of this project utilize **Google Colab Pro**.
- Propose future work:
  Image classification on Shrek characters.